The sharp four-row permanent–determinant inequality and its complete equality classification

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Abstract

For every complex 4×4 matrix A and every real $c \geq 0$, we prove $|\operatorname{per} A| + c|\det A| \leq \max\{3/2, 1+c\} \prod_i \|A_{i,*}\|_2$. The coefficient is optimal. We classify equality, including zero rows and the transition $c = 1/2$: the nondegenerate extremizers are the rank-one matrices with an equimodular common row direction, or the monomial matrices, with the relevant class selected by c. The exact real-coefficient norm of $\operatorname{per} A + t \det A$ is also determined. These principal results have complete Lean proofs for the actual matrix permanent and determinant. The written proof uses two-row identities and Laplace expansion.

1 Definitions and principal results

For $A \in \mathbb{C}^{4\times 4}$, let

$$\operatorname{per} A = \sum_{\sigma \in S_4} \prod_{i=1}^{4} A_{i,\sigma(i)}, \qquad \det A = \sum_{\sigma \in S_4} \operatorname{sgn}(\sigma) \prod_{i=1}^{4} A_{i,\sigma(i)}, \qquad R(A) = \prod_{i=1}^{4} \Big(\sum_{j=1}^{4} |A_{ij}|^2 \Big)^{1/2}.$$

A matrix is *monomial* if there is a permutation σ such that $A_{i,\sigma(i)} \neq 0$ for every i and all its other entries vanish. A nonzero *flat rank-one* matrix has the form $A_{ij} = u_i v_j$, where every u_i and v_j is nonzero and $|v_1| = \cdots = |v_4|$.

Theorem 1. *For every $A \in \mathbb{C}^{4\times 4}$ and every real $c \geq 0$,*

$$|\operatorname{per} A| + c|\det A| \leq M(c) R(A), \qquad M(c) = \max\{3/2, 1+c\}. \tag{1}$$

Equality holds if and only if at least one of the following applies:

1. *A has a zero row;*
2. *A is flat rank one and $c \leq 1/2$;*
3. *A is monomial and $c \geq 1/2$.*

The constant $M(c)$ is attained and cannot be decreased.

The permanent endpoint agrees with the sharp Hadamard-type permanent inequality of Carlen, Lieb and Loss [1]. The joint bound and its transition are proved below directly; their proof does not assume the sharp permanent bound. We make no exhaustive historical-priority claim.

Corollary 2. *For every real t,*

$$\sup_{A:\, R(A)>0} \frac{|\operatorname{per} A + t \det A|}{R(A)} = \max\{3/2, 1 + |t|\}.$$

Proof. The triangle inequality and Theorem 1, with $c = |t|$, give the upper bound. The matrix whose entries are all $1/2$ has $R = 1$, per $= 3/2$ and det $= 0$. An even permutation matrix if $t \geq 0$, and an odd permutation matrix if $t < 0$, has $R = 1$ and objective $1 + |t|$. Thus the larger endpoint is attained. □

2 The two-row calculation

For unit vectors $a, b \in \mathbb{C}^4$, define

$$S(a,b) = \sum_{j<k} |a_j b_k + a_k b_j|^2, \qquad W(a,b) = \sum_{j<k} |a_j b_k - a_k b_j|^2.$$

Write $\eta = |\sum_j a_j \overline{b_j}|^2$, $\tau = \sum_j |a_j|^2 |b_j|^2$ and $\Delta = \tau - \eta/4$. Cauchy–Schwarz gives $0 \leq \eta \leq 1$ and $\Delta \geq 0$. Expansion of the squares gives the exact identities

$$S = 1 + \eta - 2\tau, \qquad W = 1 - \eta, \qquad F_c := S + cW = 1 + c + (1/2 - c)\eta - 2\Delta. \tag{2}$$

In the second use of Cauchy–Schwarz, $\eta \leq 4\tau$ is applied to the four numbers $a_j \overline{b_j}$. Its equality condition is

$$\Delta = 0 \quad \Longleftrightarrow \quad a_j \overline{b_j} \text{ is independent of } j. \tag{3}$$

Consequently

$$0 \leq F_c(a,b) \leq M(c). \tag{4}$$

If $c < 1/2$, equality in the upper bound requires $\eta = 1$ and $\Delta = 0$. If $c > 1/2$, it requires $\eta = \Delta = 0$. At $c = 1/2$, it requires exactly $\Delta = 0$.

3 The four-row inequality and sharpness

If a row is zero, both multilinear expressions vanish, so (1) is an equality. Otherwise divide each row by its positive Euclidean norm. It suffices to prove the assertion for unit rows a, b, d, e. For a two-column subset $J = \{j, k\}$, $j < k$, put

$$p_{ab}(J) = a_j b_k + a_k b_j, \qquad w_{ab}(J) = a_j b_k - a_k b_j.$$

The permanent and determinant Laplace expansions along the first two rows are

$$\operatorname{per} A = \sum_{|J|=2} p_{ab}(J) p_{de}(J^c), \qquad \det A = \sum_{|J|=2} \varepsilon_J w_{ab}(J) w_{de}(J^c),$$

where $\varepsilon_J \in \{-1, 1\}$ is the column-shuffle sign. The complement map permutes the six two-column subsets. Hence Cauchy–Schwarz, first on these six terms and then on two real coordinates, gives

$$\begin{aligned} |\operatorname{per} A| + c|\det A| &\leq \sqrt{S(a,b)S(d,e)} + c\sqrt{W(a,b)W(d,e)} \\ &\leq \sqrt{F_c(a,b)F_c(d,e)} \leq M(c). \end{aligned} \tag{5}$$

The second step also holds at $c = 0$; for $c > 0$ it is the ordinary Cauchy–Schwarz inequality applied to $(\sqrt{S}, \sqrt{cW})$ for the two row pairs. Restoring the row norms proves (1).

The matrix with all entries $1/2$ has unit rows and objective $3/2$. Every permutation matrix has unit rows and objective $1 + c$. These examples prove optimality and attainment of $M(c)$ for every $c \geq 0$.

4 Complete equality classification

Assume all rows are nonzero and normalize them as above. Permuting rows preserves the absolute values of the permanent and determinant. Thus (5) applies to every division of the four row indices into two pairs. If its left side equals $M(c)$, both pair factors must equal $M(c)$: each is nonnegative and at most the positive number $M(c)$. Every unordered pair occurs in such a division, so all six pairs satisfy the equality conditions following (4).

If $c < 1/2$, every pair has $\eta = 1$, so all unit rows are proportional. For a pair of proportional unit rows, $\Delta = 0$ says $\sum_j |a_j|^4 = 1/4$. Since $\sum_j |a_j|^2 = 1$, equality in Cauchy–Schwarz forces $|a_j|^2 = 1/4$ for all four columns. The matrix is therefore flat rank one.

If $c > 1/2$, every pair has $\eta = \Delta = 0$, hence $\tau = 0$. The row supports are pairwise disjoint. Four nonempty pairwise disjoint subsets of four columns must each be a distinct singleton, which gives exactly a monomial matrix.

At $c = 1/2$, each pair satisfies (3). Denote its constant by z_{ih}, so

$$a_{ij}\overline{a_{hj}} = z_{ih} \quad (i \neq h) \tag{6}$$

for every column j. If the four row supports are pairwise disjoint, the same argument gives a monomial matrix. Otherwise two rows meet in one column, so their z_{ih} is nonzero and (6) forces both rows to have full support. Each remaining nonzero row meets one of these full rows and therefore also has full support. All the z_{ih} are now nonzero. Choose distinct rows i, h, k. For every column j,

$$|a_{ij}|^2 = \frac{|z_{ih}|\,|z_{ik}|}{|z_{hk}|}.$$

This is independent of j. Unit normalization makes every entry of every row have modulus $1/2$. Relative to a fixed reference row, division in (6) then makes each row a scalar multiple of that row. The matrix is flat rank one.

Conversely, for a flat rank-one matrix $A_{ij} = u_i v_j$ with common modulus $|v_j| = r > 0$,

$$|\operatorname{per} A| = 24\Big(\prod_i |u_i|\Big) r^4 = \tfrac{3}{2} R(A), \qquad \det A = 0.$$

A monomial matrix satisfies $|\operatorname{per} A| = |\det A| = R(A)$. These formulas give equality precisely at the stated weights. Together with the zero-row case this finishes the proof of Theorem 1.

5 Formal verification and scope

The repository `https://github.com/mxym/math` contains the fixed Lean projects `formalizations/four-row-permanent-tradeoff` and `formalizations/four-row-complete-equality`. At repository revision `07559d229` the latter exports `FourRowTradeoff.sharp_four_row_complete`; its matrix classes are `IsFlatRankOne` and `IsMonomial`. It covers the actual complex matrix inequality, both directions of equality, zero rows and the critical weight. The former project supplies optimality, attainment and the real-coefficient pencil norm. The complete-equality project's `FORMALIZATION_MAP.md` records the exact correspondence.

Lean is pinned to 4.34.1 (commit `5045d0056413266e57c625dcd7c365b10e377c52`); Mathlib is pinned to `d13f23b723b8a846827a245b89c10fc7d3f11612`. The archived verification completed on 9 October 2026 and reports a fresh build of all 12 owned modules and a replay of 112 named roots with a 15,593-declaration dependency closure into an empty kernel at trust level zero. The only logical axioms are `propext`, `Classical.choice` and `Quot.sound`; two invalid-proof controls were rejected. Reproduction commands and source hashes are in that project's README and `evidence/v2/VERIFICATION.json`. These are archived results; preparation of this reading edition does not claim a new Lean execution.

This paper consolidates the principal fully formalized results of the repository note `notes/four-row-permanent-tradeoff/PAPER.md`. Its additional rectangular, quantitative, convex-power and tensor extensions are outside this paper's formalization claim. No external human peer-review acceptance or mathematical-priority certification is claimed. Research and manuscript preparation used OpenAI AI assistants; the author is responsible for the stated claims. There was no external funding. Original material is all rights reserved, subject to existing third-party licenses and notices.

References

[1] E. Carlen, E. H. Lieb and M. Loss, *An inequality of Hadamard type for permanents*, Methods and Applications of Analysis **13** (2006), 1–18. doi:10.4310/MAA.2006.v13.n1.a1.

[2] Y. Zhang, *Sharp four-row permanent–determinant tradeoff and parity tensor norms*, research note and Lean source projects, 2026. `https://github.com/mxym/math/tree/07559d229/notes/four-row-permanent-tradeoff`.